

Predicting TF binding potential of DNA sequences

Course Project, Computational Functional Genomics, Jan 2026

Position weight matrices (PWMs) are traditionally used to model the in vitro sequence affinity of transcription factors (TFs). But they often fail to accurately predict actual in vivo binding maps. There are several reasons for this (can you list them?). In this project, we will integrate sequence data with in vivo chromatin maps to predict in vivo binding maps of three TFs.

Your goal is to do this for three specific TFs: CTCF, REST, and EP300 in chromosomes 3, 10, and 17 in the K562 cell-line¹. Chromosomes X and Y have been removed from the analysis. The [dataset](#) is binned at 200bp resolution across the rest of the 19 human chromosomes as separate tab separated files, one per chromosome. It has been derived from ChIP-seq peaks from ENCODE², using the hg38 genome version.

As an example, here are the first few lines of the file chr1_200bp_bins.tsv:

```
—
chr  start  end  ATAC  CTCF  REST  EP300
chr1 11800 12000 U    U    U    U
chr1 12000 12200 U    U    U    U
:
chr1 69800 70000 B    U    U    U
chr1 70000 70200 B    U    U    U
chr1 267800 268000 U    U    B    U
chr1 778600 778800 B    B    B    B
chr1 778800 779000 B    B    B    B
```

¹ This is an immortalised myelogenous leukemia cell-line

² <https://www.encodeproject.org/>. ChIP-seq peaks are considered to be gold standard for this project.

The column named “ATAC” indicates whether the chromatin at that region was open (accessible) or closed (inaccessible)³. The subsequent columns indicate binding status of CTCF, REST, and EP300, respectively. A “U” indicates the region is not bound (in case of the three TFs) or is not accessible (in case of ATAC), while a “B” indicates the opposite.

The files for the chromosomes 3, 10, and 17 contain only the first four columns. For example, here are the first few lines of chr3_200bp_bins_unknown.tsv:

```
—
chr  start  end  ATAC
chr3 12200 12400 U
chr3 12400 12600 U
chr3 13400 13600 U
:
chr3 632600      632800      B
chr3 632800      633000      B
chr3 3015400     3015600      B
```

You have to predict the probability of binding or score⁴ for each of the three TFs, where a bigger number must indicate higher binding potential. This is the score we will use to compute metrics such as receiver operating characteristic curve, precision recall curve, and the areas under them. An example of chr3_200bp_bins_predictions.tsv, i.e. output from your program could be:

```
—
chr  start  end  ATAC CTCF REST EP300
chr3 12200 12400 U    0.2  0.3  0.0
chr3 12400 12600 U    0.1  0.5  0.1
chr3 13400 13600 U    0.2  0.1  0.2
:
chr3 632600      632800      B    0.1  0.1  0.3
chr3 632800      633000      B    0.6  0.4  0.1
chr3 3015400     3015600      B    0.2  0.1  0.01
```

³ We will discuss this experiment later in the course. Until then, please ignore it.

⁴ In the case of the Markov model, this will be a log odds score, as discussed in class.

Please note that not all 200bp bins are represented in the data. Repeat regions and those with missing sequence information have been removed from both the training and prediction data. You must use an external reference genome (like the hg38.fa discussed earlier) to fetch the sequences for each bin.

Additional data

You are not allowed to use any other ChIP-seq or related high-throughput experimental data associated by TF-DNA binding, such as ChIP-exo, cut and run, etc in any cell-type. You may use other data sources if you like, for example PWM databases such as JASPAR⁵, RNA-seq data, Hi-C data, genomic features, or even evolutionary information. When in doubt, email the instructor and ask if a specific data source is okay to use.

Intermediate milestone due by midnight February 17, 2026 (via classroom)

A Markov model based classifier will be your first milestone. Each team will have a midterm evaluation based on this. Your code should take as input (a) the order m of the Markov model, (b) the number of folds k in the cross-validation, (c) tsv file for any one of the 19 training chromosomes⁶, and (d) TF name⁷. Output should be the k ROC curves, k precision-recall curves, and average area under the k ROC curves and precision-recall curves. Ensure that your program works for $m = 0$ to 10 and $k = 3$ to 5. Since the input is a single chromosome file, you should implement your k -fold split by partitioning the bins of that specific chromosome. (Ideally your code should have the ability to take data from all 19 chromosomes together, but for the intermediate goal, this will suffice.)

You may use a library to plot the curves, but the actual Markov model code (learning the parameters and scoring regions) must be yours.

⁵ <https://jaspar.elixir.no/>

⁶ This would be one of the training files, which contain 7 columns

⁷ This would be one of CTCF, REST, or EP300

You need to submit this work via a classroom Google form. The form will come online by February 15, 2026. This constitutes 10% of your final grade (accounting towards the midterm portion).

Note that for this intermediate goal, you are not using any information from the other two TFs while constructing a Markov model-based classifier for a specified TF. You are also ignoring the ATAC column. You can use the ATAC column for the final submission.

Final submission due on April 7, 2026 (via classroom)

1. Submit three separate files per team with prediction scores as described above: chr3_predictions.tsv.gz, chr10_predictions.tsv.gz, and chr17_predictions.tsv.gz.
2. You need to make a presentation towards the end of the semester. You will have to submit the slides before the end-semester exams begin. What to put in the presentation will be explained in class later.

The final submission will constitute 25% of your final grade.

Rules on collaboration

1. You may find many solutions to the original (much larger) ENCODE-DREAM challenge. However, you are **not allowed** to make use of any such publicly available code related to this challenge (the use of even small code snippets from publicly available sources related to this challenge is not permitted). You are, however, allowed to read the proposed solutions, get inspired by them, and ***develop your own solution to address the challenge.***
2. You may use other data sources if you like (see note on additional data above).
3. You may discuss strategies with others outside of your team, but **no code should be shared.**
4. If you use any additional resources, **clearly mention them** in the project presentation and also in your code (as a comment).
5. If you use AI to help you code, acknowledge the tool. Ensure that you understand each and every line of the code.

